

Jesse Burkhardt

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[Have a conversation with "me".¹](#)

EDUCATION

Ph.D. Environmental Economics, Yale University, 2016.

M.E.M. Master of Environmental Management, Yale University, 2010.

B.M. Jazz Performance, Portland State University, 2003.

EMPLOYMENT

Associate Professor, Department of Agricultural & Resource Economics, Colorado State University (2022-Present)

Assistant Professor, Department of Agricultural & Resource Economics, Colorado State University (2016-2022)

Faculty Associate, Partnership for Air Quality, Climate and Health, Colorado State University (2016-Present)

OTHER APPOINTMENTS

Editorial Council, *Journal of Environmental Economics and Management* (2023-Present)

President, W5133 Multistate Working Group: Economic Valuation and Management of Natural Resources on Public and Private Lands (2023-2024)

Pre-Doctoral Researcher, Arizona State University (2014)

Research Associate, Lawrence Berkeley National Laboratory and the Department of Energy, (2012-2014)

Advisory Board, Data Science Research Institute for high performance computing planning at CSU.

PUBLICATIONS (*GRADUATE STUDENT)

Keske, C., Hoag, D.L. and **Burkhardt, J.**, (2025). How Sense of Place Influences Willingness to Pay for Sustainable Wolf Populations. *Society & Natural Resources*, pp.1-17.

¹ Note, this can be wrong. Let me know if it comes up with anything interesting.

Wachter, Charlotte, Jordan F. Suter, **Jesse Burkhardt**, Wes Austin, Beth M. Haley, Siyu Pan, and Sonja H. Kolstoe. "Characterizing community water systems exposed to wildfire in the Western US." *Science of The Total Environment* 989 (2025): 179814.

Corwin, K.A., **Burkhardt, J.**, Corr, C.A., Stackhouse Jr, P.W., Munshi, A. and Fischer, E.V., (2025). Solar energy resource availability under extreme and historical wildfire smoke conditions. *Nature Communications*, 16(1), p.245.

Kopalle, P. K., **Burkhardt, J.**, Gillingham, K., Grewal, L. S., & Ordabayeva, N. (2024). Delivering affordable clean energy to consumers. *Journal of the Academy of Marketing Science*, 1-23.

Hill, A. E., **Burkhardt, J.**, Bayham, J., O'Dell, K., Ford, B., Fischer, E. V., & Pierce, J. R. (2024). Air pollution, weather, and agricultural worker productivity. *American Journal of Agricultural Economics*, 106(4), 1329-1353.

Burkhardt, J., Gillingham, K. T., & Kopalle, P. K. (2023). Field experimental evidence on the effect of pricing on residential electricity conservation. *Management Science*, 69(12), 7784-7798.

Maynard, R., **Burkhardt, J.**, & Quinn, J. C. (2023). Sustainability of lettuce production: A comparison of local and centralized food production. *Journal of Cleaner Production*, 428, 139224.

Hoag, D. L., **Burkhardt, J.**, Ghasemi, B., Breck, S., Niemiec, R., & Crooks, K. (2023). Willingness to pay for reintroducing wolves in a divided voting base. *Global Ecology and Conservation*, 46, e02576.

Lurbé, S., **Burkhardt, J.**, Goemans, C., Manning, D., & Hans, L. (2023). Further evidence on social comparison and residential water use. *Water Resources and Economics*, 41, 100214.

Corwin, K. A., Corr, C. A., **Burkhardt, J.**, & Fischer, E. V. (2022). Smoke-Driven Changes in Photosynthetically Active Radiation During the US Agricultural Growing Season. *Journal of Geophysical Research: Atmospheres*, e2022JD037446.

O'Dell, K., Ford, B., **Burkhardt, J.**, Magzamen, S., Anenberg, S. C., Bayham, J., ... & Pierce, J. R. (2022). Outside in: the relationship between indoor and outdoor particulate air quality during wildfire smoke events in western US cities. *Environmental Research: Health*, 1(1), 015003.

Manning, D., **Burkhardt, J.**, (2022). The Local Effects of Federal Law Enforcement Policies: Evidence from Sanctuary Jurisdictions and Crime. *Contemporary Economic Policy*. Featured in

Wiley's Research Highlights and selected as the July 2022 Featured Article on the *Contemporary Economic Policy* Website.²

Burkhardt, J., Chan, N. W., Bollinger, B., & Gillingham, K. T. (2022). Conformity and Conservation: Evidence from Home Landscaping and Water Conservation. *American Journal of Agricultural Economics*, 104(1), 228-248.

Jennings, K. S., Wlostowski, A. N., Bash, R. E., **Burkhardt, J.**, Wobus, C. W., & Aggett, G. (2022). Data availability and sector-specific frameworks restrict drought impact quantification in the Intermountain West. *Wiley Interdisciplinary Reviews (WIREs): Water*, e1586.

Bayham, J., **Burkhardt, J.**, Coffman, M., Hayashida, S., & La Croix, S. (2022). Does air pollution increase electric vehicle adoption? Evidence from US metropolitan areas, 2011–2018. *Journal of Environmental Economics and Policy*, 1-25.

Wlostowski, A.N., Jennings, K.S., Bash, R.E., **Burkhardt, J.**, Wobus, C.W. and Aggett, G., (2021). Dry landscapes and parched economies: A review of how drought impacts nonagricultural socioeconomic sectors in the US Intermountain West. *Wiley Interdisciplinary Reviews (WIREs): Water*, p.e1571.

Hill, A. E., & **Burkhardt, J.** (2021). Peers in the Field: The Role of Ability and Gender in Peer Effects among Agricultural Workers. *American Journal of Agricultural Economics*, 103(3), 790-811.

Berman, J., Bayham J., **Burkhardt, J.** (2020). Hot Under the Collar: A 14-Year Association Between Temperature and Violent Behavior across 436 U.S. Counties. *Environmental Research*.

Eldali, F., et al. (2020). A Data Decomposition Approach to Design a Dynamic Pricing Mechanism for Residence-based Plug-in Electric Vehicles in Wind Energy-rich Grids. *eTransportation*.

Bollinger, B., **Burkhardt, J.**, Gillingham, K. (2020). Peer Effects in Residential Water Conservation: Evidence from Migration. *American Economic Journal: Economic Policy*

Burkhardt, Jesse, Jude Bayham, Ander Wilson, Ellison Carter, Jesse D. Berman, Katelyn O'Dell, Bonne Ford, Emily V. Fischer, and Jeffrey R. Pierce. (2019). The effect of pollution on crime: Evidence from data on particulate matter and ozone. *Journal of Environmental Economics and Management* 98: 102267.

Burkhardt, J., and Goemans, C. (2019). The short-run effects of marijuana dispensary openings on local crime. *The Annals of Regional Science*: 1-27.

² <https://newsroom.wiley.com/press-releases/press-release-details/2022/Sanctuary-city-policies-linked-to-lower-crime-rates-in-recent-years/default.aspx>

Chan, N. W., **Burkhardt, J.** & Flyr, M. (2019). The effects of recreational marijuana legalization and dispensing on opioid mortality. *Economic Inquiry*. Featured in Wiley's Research Highlights

Burkhardt, J., et al. (2019). The relationship between monthly air pollution and violent crime across the United States." *Journal of Environmental Economics and Policy*: 1-18.

Berman, J. D., **Burkhardt, J.**, Bayham, J., Carter, E., & Wilson, A. (2019). Acute air pollution exposure and the risk of violent behavior in the United States. *Epidemiology*, 30(6), 799-806. *This paper was awarded "Best Environmental Epidemiology Paper published in 2019," by the International Society for Environmental Epidemiology.*

Burkhardt, J. (2019). The impact of the Renewable Fuel Standard on US oil refineries. *Energy Policy*, 130, 429-437.

Maas, A., Goemans, C., Manning, D. T., **Burkhardt, J.**, & Arabi, M. (2019). Compliments of the house: Estimating demand-side linkages between residential water and electricity. *Water Resources and Economics*, 100140.

Flyr, M.,* **Burkhardt, J.**, Goemans, C., Hans, L., Neel, A., & Maas, A. (2019). Modeling Commercial Demand for Water: Exploring Alternative Prices, Instrumental Variables, and Heterogeneity. *Land Economics*, 95(2), 211-224.

Burkhardt, J. and Flyr, M.* (2018). The Effect of Marijuana Dispensary Openings On Housing Prices. *Contemporary Economic Policy*. doi:10.1111/coep.12414

Burkhardt, J. and Chan, N.W. (2017). The dollars and sense of ballot propositions: estimating willingness to pay for public goods using aggregate voting data. *Journal of the Association of Environmental and Resource Economists* 4.2: 479-503.

Burkhardt, J., Wiser, R., Darghouth, N., Dong, C. G., & Huneycutt, J. (2015). Exploring the impact of permitting and local regulatory processes on residential solar prices in the United States. *Energy Policy*, 78: 102-112.

WORKING PAPERS

Bias Induced by Fixed Effects in the Presence of Collider Variables

Tradeoffs of Different Measurements of Wildfire Smoke and other Air Pollutants

NON-PEER REVIEWED PUBLICATIONS

Burkhardt, Jesse., Gifford, Lauren. (2023). Climate change is about to play a big role in government purchases – with vast implications for the US economy. *The Conversation*. <https://theconversation.com/climate-change-is-about-to-play-a-big-role-in-government-purchases-with-vast-implications-for-the-us-economy-214549>

Hoag, Dana., Burkhardt, Jesse., Ghasemi, Ben., Young, Robin. (2023). Economic wins and losses from reintroducing wolves in Colorado. *REDI Report*. <https://csuredi.org/>.

Anne, C. K., Chelsea, C., & Jesse, B. (2022). Smoke-Driven Changes in Photosynthetically Active Radiation During the US Agricultural Growing Season. *Authorea Preprints*.

Corwin, K., Corr, C., **Burkhardt, J.**, & Fischer, E. V. (2022, December). Historical Exposure of US Solar Energy Potential to Smoke-Driven Changes in Shortwave Irradiance. In *Fall Meeting 2022*. AGU.

Corwin, K., Fischer, E., Corr, C., & **Burkhardt, J.** (2021, December). Fires to Farms: How Wildfire Smoke Impacts Surface-Level Photosynthetically Active Radiation. In *AGU Fall Meeting Abstracts* (Vol. 2021, pp. GC42B-07).

Goemans, C., Manning, D., **Burkhardt, J.** (2021). An Analysis of the Impact of Drought on Agriculture, Local Economies, Public Health, and Crime Across the Western United States, *NOAA National Integrated Drought Monitoring System and the Colorado Water Conservation Board*.

O'Dell, K., Ford, B., **Burkhardt, J.**, Anenberg, S., Bayham, J., Fischer, E., & Pierce, J. (2021, December). Influence of wildfire smoke on indoor air quality in several western US cities. In *AGU Fall Meeting Abstracts* (Vol. 2021, pp. A33G-02).

Fischer, E. V., Pierce, J. R., Magzamen, S., Ford, B., O'Dell, K., Brey, S. J., ... & Abdo, M. (2020, December). Enabling Smoke Impacts Research Through Blended Earth Observations. In *AGU Fall Meeting Abstracts* (Vol. 2020, pp. GH016-01).

Burkhardt, J., Gillingham, K., Kopalle, P.K. (2019). Experimental Evidence on the Effect of Information and Pricing on Residential Electricity Consumption, *NBER Working Paper*, <https://www.nber.org/papers/w25576.pdf>

Burkhardt, J., Gillingham, K., Kopalle, P.K. (2019). Experimental Evidence on the Effect of Information and Pricing on Residential Electricity Consumption. SSRN Working Paper: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3336385

Bollinger, B., **Burkhardt, J.**, Gillingham, K. (2018). Peer Effects in Water Conservation: Evidence from Consumer Migration, *NBER Working Paper*, <http://www.nber.org/papers/w24812>

Burkhardt, J., Bollinger, B., Gillingham, K. (2018). Peer Effects in Water Conservation: Evidence from Consumer Migration, *Global Water Forum*, <http://www.globalwaterforum.org/2018/08/13/peer-effects-in-water-conservation-evidence-from-consumer-migration/>

Burkhardt, J., Kopalle, P., Gillingham, K. (2018). How do Residential Consumers Respond to Critical Peak Pricing? Experimental Evidence on the Role of Information and Incentives, Extended Abstract. *Association for Consumer Research*, <https://convention2.allacademic.com/one/acr/acr18/>

Modelling Commercial Demand for Water, Colorado Water, Sept/Oct 2018.
https://issuu.com/coloradowater/docs/cw_35_5?e=1964603/66317052

Faculty Profile, Colorado Water, Sept/Oct 2018.
https://issuu.com/coloradowater/docs/cw_35_5?e=1964603/66317052

GRANT AND FELLOWSHIP ACTIVITY

NOAA Cooperative Agreement (2024-Present). PI. Data Engineering for Weather Forecasting Errors. **Funded.**

USDA NIFA. (2025). PI. Combining Earth Observations with Machine Learning to Detect Agricultural Methane Emissions. \$650,000. **Pending.**

USDA NIFA. (2025). PI. Estimating US CAFO Ammonia Emissions Using Satellite and Ground-Based Observations for Economic, Health, and Policy Analysis. \$650,000. **Pending.**

Sloan Foundation. (2025). PI. Combining Machine Learning and Satellite Observations to Estimate Methane Emissions and Associated Energy Losses from Concentrated Animal Feeding Operations. LOI. **Not Funded.**

Burrows Fund. (2025). Co-PI. Rocky Mountain Center for Research, Education, and Action on Climate and Health (RM-REACH). LOI. **Not Funded.**

NASA. (2025). PI. Rapid Response Earth Systems. Combining State-of-the-art Machine Learning and Satellite Observations to Estimate Methane Emissions from Concentrated Animal Feeding Operations. **Not Funded.**

School of Global Environmental Sustainability: Global Challenges Research Team Award. (2025-2027). PI. Combining State-of-the-art Machine Learning and Satellite Observations to Estimate Methane Emissions from Concentrated Animal Feeding Operations. \$30,000. **Not Funded.**

NASA. (2025). PI Advisor. F.5 Future Investigators in NASA Earth and Space Science and Technology. \$150,000. **Funded.**

NIH. (2024). Co-PI. Supporting Faculty. MAP ERC Targeted Research Training program on Climate & Worker Health. **Not Funded.**

NASA. (2024). PI. Leveraging Satellites and Machine Learning to Classify Global Wildfire Smoke Presence and Exposure \$1,112,597. **Not Funded.**

NASA. (2024). PI. Leveraging Satellites and Machine Learning to Classify Global Wildfire Smoke Presence and Exposure \$1,045,397. **Not Funded.**

NASA. (2024). Co-PI. Leveraging satellites with regulatory and low-cost monitors to facilitate air pollution decision making and health research. **Not Funded.**

Data Science Research Institute. (2024). PI. Predicting Global Wildfire Smoke Exposure Using AI. \$28,000. **Funded.**

School of Global Environmental Sustainability: Global Challenges Research Team Award. (2024-2026). Co-PI. Global Climate Change Related Wildfire Damages. \$30,000. **Funded.**

Colorado State Government (2023). Co-PI. Developing the socio-economic and dynamic Life Cycle frameworks for Agrivoltaics adoption in Colorado. \$200,000. **Not Funded.**

USDA Economic Research Service (2022-2024). PI. Towards the Development of a National Agricultural Methane Monitoring System and the Socio-Economic Implications of Methane Regulation. \$284,537. **Not Funded.**

USDA Economic Research Service (2022-2024). PI. Understanding the Social and Economic Barriers to Large Scale Adoption of Agrivoltaics in Colorado. \$289,600. **Not Funded.**

National Aeronautics and Space Administration (NASA) (2023-2025). Co-PI. Leveraging satellites to understand to impacts of air pollution on farm-worker productivity and wellbeing. \$1,305,273. **Not Funded.**

USDA Forest Service Cooperative Agreement (2022-2025). Co-PI. A Comparative Analysis of Wildfire Impacts on Western Drinking Water Systems. \$80,400. **Funded.**

National Park Service Cooperative Agreement (2022-2024). PI. Non-Market Valuation of National Parks Cooperative Agreement. \$149,660. **Funded.**

Department of Energy Foundational Agrivoltaic Research for Megawatt Scale (FARMS) (2023-2025). PI. Understanding the Social, Economic, and Regulatory Barriers to Large Scale Adoption of Agrivoltaics. \$1,732,272. **Not Funded.**

NCAR Early Career Faculty Innovator Program (2022-2025). PI. Towards the Development of a National Agricultural Methane Emissions Monitoring System. \$200,000. **Not Funded.**

EPA-G2022-STAR-F1, Drivers and Environmental Impacts of Energy Transitions in Underserved Communities (2023-2025). Co-PI. Colorado Household Equity and EneRgy Study (CHEERS) – Adoption, Impacts, and Policy Levers. \$500,000. **Not Funded.**

Foundation for Food and Agricultural Research (FFAR). (2022-2025). Co-PI. Agrivoltaics Supplying Urban Food Systems, Willingness to Pay, and Food/Energy System Modeling Impacts. \$1,374,703.66. **Not Funded.**

School of Global Environmental Sustainability: Global Challenges Research Team Award. (2022-2024). PI. Wildfire Smoke Impacts on U.S. Solar and Agricultural Productivity. \$30,000. **Funded.**

USDA-NIFA-National Institute of Food and Agriculture. 2021. Co-PI. Fires to Farms: How do Wildfire Smoke Driven Changes in Radiation Impact Crops. \$642,968.80. **Not Funded.**

US EPA-OW-OGWDW-20-02, Office of Water, Office of Ground Water and Drinking Water, 2021. Co-PI. Towards Achieving SUCCESS in Drinking Water. \$2,376,218. **Not Funded.**

USDA-NIFA-National Institute of Food and Agriculture. 2020. Co-PI. Fires to Farms: How do Wildfire Smoke Driven Changes in Radiation Impact Crops. \$499,764.58. **Not Funded.**

NOAA Climate Program Office. NIDIS: Coping with Drought. 2019-2022. PI. Developing Multi-Sector Drought Impact Models for the Intermountain West DEWS: Towards an Impact-and-Data-Driven Approach for Drought Management. \$458,232 (\$62,244.76 to CSU). **Funded.**

NOAA NIDIS RFP. 2019-2021. Co-PI. Developing drought impact models for the Intermountain West Drought Early Warning System. \$650,000. **Funded.**

USDA-NIFA-National Institute of Food and Agriculture. 2019. Co-PI. Fires to Farms: How do Wildfire Smoke Driven Changes in Radiation Impact Crops. \$479,213.47. **Not Funded.**

SoGES 2018-19 Global Challenges Research Team Proposal. 2018-2019. Co-PI. Sustainable Performance of Healthy and Efficient Residential Environments (SPHERE) \$20,000. **Funded.**

National Science Foundation Energy Research Centers Grant. 2019. Co-PI. **Not Funded.**

Graduate School Student Recruitment Mini-Grant. 2018. Co-PI. **Funded.**

SoGES 2018-19 Global Challenges Research Team Proposal. 2018-2019. Co-PI. Next Generation Electricity Demand Response. \$20,000. **Funded.**

National Science Foundation. 2018-2021. Co-PI. Spatio-temporal coordination for increased utilization of renewable resources in the electric grid. \$700,000. **Not Funded.**

U.S. Department of Agriculture, Forest Service and Colorado State University Vice President of Research Seed Grant, 2016-2017. PI. Assessing the Global Economic Impacts of Predicted Fishery Decline. \$29,070. **Funded.**

Colorado State University Water Center Fellow, 2017. PI. From Information to Prices: What Drives Residential and Commercial Water Demand? \$10,000. **Funded.**

Environmental Protection Agency. 2017-2020. Co-PI. Forecasting the Heterogeneous Adoption and Environmental Impacts of Clean Energy Technology in the United States. \$300,000. **Not Funded.**

National Science Foundation. 2017-2020. Co-PI. Spatiotemporal Variability in Clean Energy Technology Uptake and Environmental Impacts under a Decarbonizing Future. \$279,895.22. **Not Funded.**

National Science Foundation. 2017-2020. Co-PI. CPS: Medium: Collaborative Research: Spatio-Temporal Coordination for Increased Utilization of Renewable Resources in the Electric Grid. \$700,000. **Not Funded.**

National Science Foundation, 2017-2021. Co-PI. INFEWS/T3: Resource-Efficient Thin-Film Photobioreactor System for Fuel and Fertilizer Production. \$2,500,000. **Not Funded.**

SoGES 2017-18 Global Challenges Research Team Proposal. 2017. Co-PI. Next Generation Electricity Demand Response. \$20,000. **Not Funded.**

Arizona State University Pre-Doctoral Fellowship in Environmental and Resource Economics. 2013. **Graduate Salary.**

Yale Institute for Biospheric Studies, Dissertation Improvement Grant. 2013. \$5,000.

National Science Foundation Integrative Graduate Education and Research Traineeship Grant. 2011. (Declined to attend Yale)

Doris Duke Conservation Fellowship. 2009-2010. \$70,000.

American Alpine Club and Mountain Hardware grant to establish new climbing routes in Tajikistan. 2010.

INVITED PRESENTATIONS

Citizens Climate Lobby, *Introduction to Climate Economics*, (Fall, 2024)

NASA Air Quality and Health Applied Sciences Team Environmental Justice Meeting (Spring, 2024). *Air Pollution, Weather, and Agricultural Worker Productivity*

University of Wyoming Economics Seminar (Spring, 2024)

SOGES Panel: The Renewable Energy Transition and what it means for Colorado (Fall, 2023)

SOGES Panel: The Renewable Energy Transition and what it means for Colorado (Fall, 2022)

Northern Colorado Clean Cities: *The economics of electric vehicles and where the power comes from* (February, 2021)

Agricultural Impact Drought Task Force: *The impact of drought on Western Agriculture and other Socioeconomic Sectors*. (November, 2020)

Colorado State University Water Center: *Water Center Fellow Report* (May, 2019)

University of Wyoming Seminar to Economics Club (Spring, 2019). *Three papers on the socioeconomic effects of marijuana legalization*

Colorado School of Mines Economics Seminar (February, 2018). *Peer Effects in Dry Landscape Adoption*

University of Wyoming Economics Seminar (Fall, 2018). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

University of California Los Angeles Industrial Organization Workshop (Spring, 2017). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

New York University Industrial Organization Seminar (Fall, 2017). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

Colorado State University Economics Seminar (Fall, 2016). *Peer Effects in Dry Landscape Adoption*

Arizona State University Urban Water Demand Roundtable (Fall, 2016). *Peer Effects in Dry Landscape Adoption*

Yale University Environmental Economics Seminar (Fall, 2015). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

Arizona State University Urban Water Demand Roundtable (Fall, 2015). *Peer Effects in Dry Landscape Adoption*

CONFERENCES

Fitzgerald et al. (2026). Economic Consequences of Wildfire Smoke Avoidance Behavior. ASSA Conference. Presented by graduate student.

Fitzgerald et al. (2025). Economic Consequences of Wildfire Smoke Avoidance Behavior. AERE Conference. Presented by graduate student.

Fitzgerald et al. (2025). Predicting global smoke exposure using machine learning and Earth observations. TWEEDS Conference. Presented by graduate student.

Duhal et al., (2025). Willingness to Participate in Agricultural Water Conservation Programs: Choice Experiment Evidence from the Upper Colorado River Basin. AAEE Annual Meeting. Presented by graduate student.

Corwin, K., Corr, C., Burkhardt, J., & Fischer, E.V. (2024). *Solar energy resource availability under extreme and historical wildfire smoke conditions*. AGU Fall Meeting 2024. **Won 1st Place for Student 15Energy Competition.**

Wachter, Charlotte., Suter, J., Burkhardt, J., et al., (2024). *Characterizing Community Drinking Water Systems Exposed to Wildfire Impacts in the Western U.S.* Western Agricultural Economics Association Annual Meeting.

Corwin, K., Corr, C., Burkhardt, J., & Fischer, E.V. (2022, December 12-16). *Historical Exposure of U.S. Solar Energy Potential to Smoke-Driven Changes in Shortwave Irradiance* [Poster presentation]. AGU Fall Meeting 2022, Chicago, IL, United States. <https://ui.adsabs.harvard.edu/abs/2022AGUFMGC22K0707C>

Corwin, K., Corr, C., Burkhardt, J., & Fischer, E.V. (2023, January 8-12). *Historical Exposure of U.S. Solar Energy Potential to Smoke-Driven Changes in Shortwave Irradiance* [Poster presentation]. AMS 103rd Annual Meeting, 14th Conference on Weather, Climate, and the New Energy Economy, Denver, CO, United States.

American Geophysical Union (AGU) Annual Meeting (2022). *Historical Exposure of U.S. Solar Energy Potential to Smoke-Driven Changes in Shortwave Irradiance*. Presented by Kimberley Corwin (Graduate Student).
<https://agu.confex.com/agu/fm22/prelim.cgi/Paper/1052303>

American Geophysical Union (AGU) Annual Meeting (2021). *Fires to Farms: How Wildfire Smoke Impacts Surface-Level Photosynthetically Active Radiation*. Presented by Kimberley Corwin (Graduate Student).

Workshop in Energy and Environmental Research: University of Hawaii. (2021) *Does Air Pollution Increase Electric Vehicle Adoption? Evidence from U.S. Metropolitan Areas, 2011-2018*.

W4133 (March 2022). Attended as Secretary.

POWER (March 2021). Attended.

W4133 (March 2021). Attended.

WEAI Annual Conference (2020). *The impact of pollution exposure on agricultural worker productivity*. (Conference Cancelled due to Covid 19).

W4133 (February 2020). *Do people respond to pollution and wildfire smoke? Evidence from the American Time Use Survey*. Presented by Coauthor

American Water Works Association Annual Conference (June, 2019). *Consumer Responses to Social Comparison* (presented by graduate student)

AERE (May 2019). *The Effect of Pollution on Aggressive Behavior*. Presented by Coauthor

Front Range Energy Economics Workshop (May 2019). *Do people respond to pollution and wildfire smoke? Evidence from the American Time Use Survey*

Front Range Energy Economics Workshop (May 2019). *Welfare Effects of Electricity Grid Shutdowns* (presented by coauthor)

W4133 (February 2019). *The Effect of Pollution on Aggressive Behavior*.

ASSA Annual Meeting (January 2019). *Willingness to Pay for Conformity (presented by coauthor)*

AERE at Southern Economics Association Conference (November 2018). *The Effect of Pollution on Aggressive Behavior*

UC Boulder Environmental Economics Workshop in Vail (August 2018). *Willingness to Pay for Conformity (presented by coauthor)*

AAEA Annual Meeting (September 2018). *Consumer Responses to Social Comparison (presented by graduate student)*

Western Agricultural Economics Association (June 2018). *Estimating Heterogeneity in the Price Elasticity of Commercial Water Users (presented by graduate student)*

Colorado State University Graduate Student Symposium (April 2018). *Consumer Responses to Social Comparison (presented by graduate student)*

Colorado State University Graduate Student Symposium (April 2018). *Estimating Heterogeneity in the Price Elasticity of Commercial Water Users (presented by graduate student)*

Front Range Energy Economics Workshop (May 2018). *How do Residential Consumers Respond to Critical Peak Pricing?*

Front Range Energy Economics Workshop (May 2018). *The effect of pollution on crime (presented by Jude Bayham)*

National Bureau of Economic Research (NBER) Energy and Environmental Economics Spring Session (March 2018). *Peer Effects in Dry Landscape Adoption*

NBER Workshop on Commodity Prices (Fall 2016).

NBER Summer Institute for Environmental and Energy Economics (Spring 2015). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

CAMP Resources (Spring 2015). *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

Northeast Energy Policy and Environmental Economics Workshop at Yale (Spring 2015) *Incomplete Regulation of an Imperfectly Competitive Market: The Impact of the Renewable Fuel Standard on U.S. Oil Refineries*

Arizona State Department of Economics (Spring 2014) *Peer Effects in Dry Landscape Adoption*

Yale University Graduate Student Conference (2013)

George Wright Society Conference on Parks, Protected Areas, and Cultural Sites (2011)

COURSES TAUGHT

Graduate Machine Learning for Econometrics (Fall, 2024-Present)
Graduate Environmental Economic (Spring, 2024-Present)
Graduate Econometrics: Causal Inference (Fall, 2022-Present)
Introduction to Econometrics (Spring, 2016-2023)
Economics of Energy Resources (Spring, 2016-Present)
Introduction to Econometrics (Online Spring, 2021)
Economics of Energy Resources (Online Spring, 2021)
Interdisciplinary Graduate Seminar on Integrated Assessment Models and the Social Cost of Carbon (Fall, 2017)

GUEST LECTURES

Graduate Research Methods, (Fall, 2024)
Shared Socioeconomic Pathways, Atmospheric Sciences (Fall, 2023-24)
Global Environmental Sustainability (GES) 141 (Fall, 2020)
Masters Level Course on the Social Cost of Carbon (Fall, 2019)
Masters Econometrics (Fall, 2018, 2019)
Honors Seminar in Conflict Management (Fall, 2017)
Engineering Economics in Mechanical Engineering (Spring, 2018, 2019, 2020, 2021)
Climate Change in Atmospheric Sciences (Spring, 2018)
Introduction to Petroleum Geology (Spring, 2018)

TEACHING FELLOWSHIPS

Economics of the Environment (2014-2015) Yale University
Energy Economics and Policy Analysis, (2015), Yale University

REFeree SERVICE

American Economic Journal: Economic Policy; Journal of Policy Analysis and Management; Journal of Economics and Human Biology; Economic Inquiry; Contemporary Economic Policy; Energy Economics; Environmental and Resource Economics; Journal of Crime and Delinquency; Journal of Environmental Economics and Management; Journal of the Association of Environmental and Resource Economists; Energy Policy; American Journal of Agricultural Economics; Current Science; Agricultural and Applied Economics Association (AAEA); Resource and Energy Economics; PLOS One; Land Economics; Southern Economics Association Annual Conference; Nature Energy; Journal of Environmental Management; Sloan Foundation; Journal of Agricultural and Resource Economics;

PROFESSIONAL SERVICE AND TRAININGS

hfp Consulting Leadership Training (2024)
Inclusive Pedagogy Workshop II (Nov 30, 2022)
IDEA director search committee (2022-2023)

Chair of DARE Curriculum (2022-present)
Member of the Colorado Climate Change Technical Advisory Group (2022)
Designing Your Online Course (CSU) (2021)
Search Chair Training (CSU) (June 2021)
Attend and participate in monthly Colorado Electric Vehicle Coalition meetings and subgroup meetings (2020-2022)
Secretary for W4133 (2021)
Design Center for Community Resilience & Well-Being Working Group (2021)
CAS Strategic Planning Committee (2020)
Faculty Council, Colorado State University (2019-Present)
Undergraduate Committee, Colorado State University (2017-Present)
Environment and Resource Economics Undergraduate Majors Club Faculty Representative (2019-2022)
College of Agricultural Sciences Academic Affairs Committee (Member 2021)
Graduate Placement Coordinator, Colorado State University (2017-Present)
Environmental and Resource Economics Lab Organizer (2018-present)
Faculty Search Committee, Colorado State University (2016-17)

MEDIA

[Wildfire smoke causes only ‘modest reduction’ in solar power — study.](#) (2025). *Politico*

[Research explores wildfire smoke’s effect on solar power generation across U.S.](#) Source. (2025)

[Supply Chain Tradeoffs are Never Simple.](#) Source. (2023)

[Potentially Good News for Solar Energy During Wildfires.](#) *EOS*. (18 January 2023).

Wiley Research Headlines. (2022) WEAI nominated and selected article “The Local Effects of Federal Law Enforcement Policies: Evidence from Sanctuary Jurisdictions and Crime,” presented on Wiley website.

Drought Impacts on the Western US. Interview with Judith Kohler from the Denver Post. (2021).

Air Pollution Impact on Domestic Violence: Fuller Project for International Reporting. (2020).

[A Correction:](#) Podcast Interview. (2019)

[Study says air pollution can lead to violence in U.S.](#) NBC News. (2019)

[CSU study supports link between air pollution and violent crime.](#) *The Coloradan*. (2019)

[Air pollution can lead to violent crimes, says American research.](#) *Hindustan Times*. (2019)

[On edge? It might be what’s in the air.](#) *Earth Day Network*. (2019)

[Exposure to air pollution is linked to an increase in violent crime.](#) *The Economist*. (2019)

Kentucky News Connection. Public News Service. (2019)

[Air pollution and violence linked.](#) *Duluth News Tribune*. (2019)

[Air pollution exposure linked to aggressive behaviour: US study.](#) *Business Standard*. (2019)

[Exposure to Air Pollution Can Make You More Aggressive: Research.](#) *Weather.com* (2019)

[Exposure to air pollution increases violent crime, study suggests.](#) *Air Quality News.com*. (2019)

[Exposure to air pollution increases violent crime rates.](#) *Science Daily*. (2019)

[New Study Links Air Pollution and Violent Crime.](#) *Our Daily Planet*: (2019)

[Air pollution may affect crime, but the problem goes deeper than that.](#) *Gizmodo* (2019)

Exposure to air pollution increases violent crime rates, study finds. *Carleton University's science magazine Catalyst* (2019)

Colorado State University News. (2019) [Exposure to air pollution increases violent crime rates, study finds.](#)

Fox News. (2019) [Air Pollution Linked to Violent Behavior Says University of Minnesota Study.](#)

Daily Mail. (2019) The Effects of Recreational Marijuana Legalization on Opioid Mortality.

Science Daily: (2019) [Marijuana Legalization Reduces Opioid Deaths.](#)

Agence France-Presse. (2019) The Effects of Recreational Marijuana Legalization on Opioid Mortality.

Wiley Research Headlines. (2019) WEAI nominated and selected article “The Effects of Recreational Marijuana Legalization on Opioid Mortality,” presented on Wiley website.

Dartmouth Tuck School of Business News: [The Key to Energy Conservation? Priming Consumers with Price.](#) (2019)

NBER Featured Working Paper on Front Page of NBER Website: Experimental Evidence on the Effect of Information and Pricing on Residential Electricity Consumption (2019)

Cited by Boulder Weekly. “[For yuppies, pot is ceasing to be a guilty pleasure.](#)” By Paul Danish December 6, 2018.

National Public Radio (Spring, 2017) Audio news article about interdisciplinary seminar on social cost of carbon.

OUTREACH AND COMMUNITY

Participated in Electric Vehicle Symposium for the Northern Colorado Clean Cities (2021)

Served as an expert on the Renewable Fuel Standard for the US Environmental Protection Agency and the US Government Accountability Office (2020)

Secured internships for two graduate students at the Fort Collins Utility that led to presentations to the utility, data services for the utility, and two publications
Participated in Interdisciplinary Workshop hosted by SOGES (2018)
Participated in Collaborative Workshop between CSU and Fort Collins Utilities (2018)
Presentation to the American Lung Association (2017), Research Summary and Opportunities for Collaboration (RSC)
Presentation to the Colorado Energy Office (2017), RSC
Presentation to the Colorado Electric Vehicle Coalition (2017), RSC

STUDENTS

Undergraduate Students

Justin Smith (2018)
Jenna Stubbers (2021-2022)
Mia Marones (2022)
Kate Phillips (2023-2024)

Masters Students

Evan Stockmoe (Chair, 2023-2024)
Matthew Flyr (Chair, 2016-2018)
Jesse Cruce (Outside Member, 2017-2019)
Mingsui Sun (Outside Member, 2017-2019)
Derek Young (Outside Member, 2017-2018)
Gal Koss (Secured Summer Funding for Them, 2020)
Luc Lallement (Chair, 2022-present)
Evan Stockmoe (Chair, 2022-present)
Gavin LeGrange (Outside Member, 2022-present)
Natalia Goncharova (Outside Member, 2023)

Doctoral Students

Brooke Fitzgerald (Chair, 2023-Present)
Charlotte Watcher (Co-Chair, 2023-Present)
Jaeyong Jung (Chair, 2024-Present)
Henry Williams (Outside Committee Member, 2023-Present)
Julius Abayateye (Outside Committee Member, 2023-2025)
Jacob Mdigo (Outside Committee Member, 2023-2025)
Karissa Palmer (Outside Committee Member, 2023-Present)
Shreezal GC (Committee Member, 2025-Present)
James Larson (Outside Committee Member, 2025-Present)
Luke Maddock (Outside Committee Member, 2024-Present)

Past Doctoral Students

Kimberley Corwin (Outside Committee Member and InTERFEWS Mentor, 2020-2024)
Reid Maynard (Outside Committee Member, 2020-2022)
Zarif Rasul (Chair, 2023-2024)

Salvador Lurbe (Committee Member, 2016-2021)
Annabelle Berklund (Outside Member, 2017-2019)
Putri Dara (Advised and Employed Via State Grant, 2019-2020)
Siddharth Kishore (Advised and Employed Via State Grant, 2019-2020)
Paige Nankey (Outside Committee Member, 2021-2022)
Ben Parker Platt (Outside Committee Member, 2021)
Jorgen Rasmussen (Outside Committee Member, 2021-2022)
Ben Saarloos (Outside Committee Member, 2021-2022)
Victoria Omojoso (Committee Member, 2021-2022)
Shuiqin Yu (Committee Member, 2019-Present)

AWARDS

College of Agriculture CSU Applied Policy Research Award (2024)
AAEA Bruce Gardner Award (2024)
Manning and Burkhardt (2022) Featured in Wiley's Research Highlights and selected as the July 2022 Featured Article on the *Contemporary Economic Policy* Website.
Paper awarded "Best Environmental Epidemiology Paper published in 2019," by the International Society for Environmental Epidemiology.
Faculty Appreciation from Agricultural Fraternity at CSU
Best graduate student presentations at CSU graduate student symposium by Matthew Flyr (Masters Student) and Salvador Lurbe (Ph.D. Student).

OTHER PROFESSIONAL EXPERIENCE

Research Assistant, Institute of Transportation Studies, University of California Davis (2012)
Research Assistant, Dr. Matthew Kotchen and Dr. Kory Kroft, Yale University (2010-2012)
Backcountry Wilderness Ranger, Yosemite National Park, California (2004-2007)
Goat Cheese Maker, Les Courmettes, France (2005)

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